



2.5 Modelling with technology and model fitting

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Technology (GDC/software) can fit regression models — linear, quadratic, cubic, exponential, or power — to bivariate data. The coefficient of determination R^2 ($0 \leq R^2 \leq 1$) measures the proportion of variability in the data explained by the model; a value closer to 1 indicates a better fit. The best model is chosen by combining a high R^2 with contextual sense, including how the model behaves under extrapolation.

Worked example: follow the reasoning

A scientist fits two models to temperature (x) vs reaction rate (y) data: a linear model ($R^2=0.87$) and an exponential model ($R^2=0.98$). Which model should be chosen, and why?

Solution: Choose the exponential model, since $R^2=0.98$ is closer to 1 than 0.87, meaning it explains more of the variability in the data; this also matches the way reaction rates often increase multiplicatively with temperature rather than by a constant amount.

Quick check · multiple choice

What does an R^2 value of 0.95 for a fitted regression model indicate?

- A 95% of the data points are exactly correct
- B The model explains about 95% of the variability in the data
- C There is a 95% chance the model is correct
- D The correlation coefficient is 0.95^2

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A researcher records the population, P thousand, of a bacteria colony against time t hours, and fits a quadratic model $P=0.5t^2+2t+10$ ($R^2=0.89$) and an exponential model $P=10(1.15)^t$ ($R^2=0.99$) using technology. (a) State which model is the better fit, justifying your answer using R^2 . (b) Use the exponential model to predict the population at $t=15$ hours, to 1 decimal place. (c) Explain why the exponential model is generally more appropriate than a quadratic model for bacterial growth. [7]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) The exponential model, since $R^2=0.99$ is closer to 1 than 0.89, so it explains a greater proportion of the variability in the data. (b) $P(15)=10(1.15)^{15} \approx 10 \times 8.14 \approx 81.4$ thousand. (c) While resources are unlimited, bacterial growth rate is proportional to the current population size, which produces multiplicative (exponential) growth; a quadratic model has no such biological basis and would eventually diverge from real behaviour.

Quick check answer

B. R^2 is the proportion of the variation in the dependent variable accounted for by the model.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.